

TACC-Hadoop

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Get your hands on

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Play-on Guide

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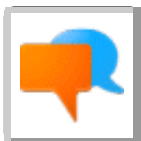
Trouble Shooting

- When you cannot upload files to HDFS
- [Fatal Error] core-site.xml:1:1: Premature end of file
- Error occurred during initialization of VM
- INFO ipc.Client: Retrying connect to server: c201-116/129.114.19.21:9000. Already tried 0 time(s).

Files

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Word Count Example Part I - Create Your Own Jar

- open Eclipse
- Create a project and a class. The example class is named "WordCount".
- Get the java code for word count [here](#).

```
package org.myorg;
```

```
import java.io.IOException;
import java.util.*;
```

```
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.conf.*;
import org.apache.hadoop.io.*;
import
org.apache.hadoop.mapreduce.*;
import
org.apache.hadoop.mapreduce.lib.in
put.FileInputFormat;
import
org.apache.hadoop.mapreduce.lib.in
put.TextInputFormat;
import
org.apache.hadoop.mapreduce.lib.ou
tput.FileOutputFormat;
import
org.apache.hadoop.mapreduce.lib.ou
tput.TextOutputFormat;
```

```
public class WordCount {
```

```
    public static class Map extends
Mapper<LongWritable, Text, Text,
IntWritable> {
        private final static
IntWritable one = new
IntWritable(1);
        private Text word = new
Text();
```

```

        public void map(LongWritable
key, Text value, Context context)
throws IOException,
InterruptedException {
            String line =
value.toString();
            StringTokenizer tokenizer
= new StringTokenizer(line);
            while
(tokenizer.hasMoreTokens()) {

word.set(tokenizer.nextToken());
                context.write(word,
one);
            }
        }

    public static class Reduce
extends Reducer<Text, IntWritable,
Text, IntWritable> {

        public void reduce(Text key,
Iterator<IntWritable> values,
Context context)
            throws IOException,
InterruptedException {
                int sum = 0;
                while (values.hasNext()) {
                    sum +=
values.next().get();
                }
                context.write(key, new
IntWritable(sum));
            }
        }

    public static void main(String[]
args) throws Exception {
        Configuration conf = new
Configuration();

        Job job = new Job(conf,
"wordcount");

job.setOutputKeyClass(Text.class);

job.setOutputValueClass(IntWritabl
e.class);

        job.setMapperClass(Map.class);

        job.setReducerClass(Reduce.class);

```

```
job.setInputFormatClass(TextInputFormat.class);

job.setOutputFormatClass(TextOutputFormat.class);

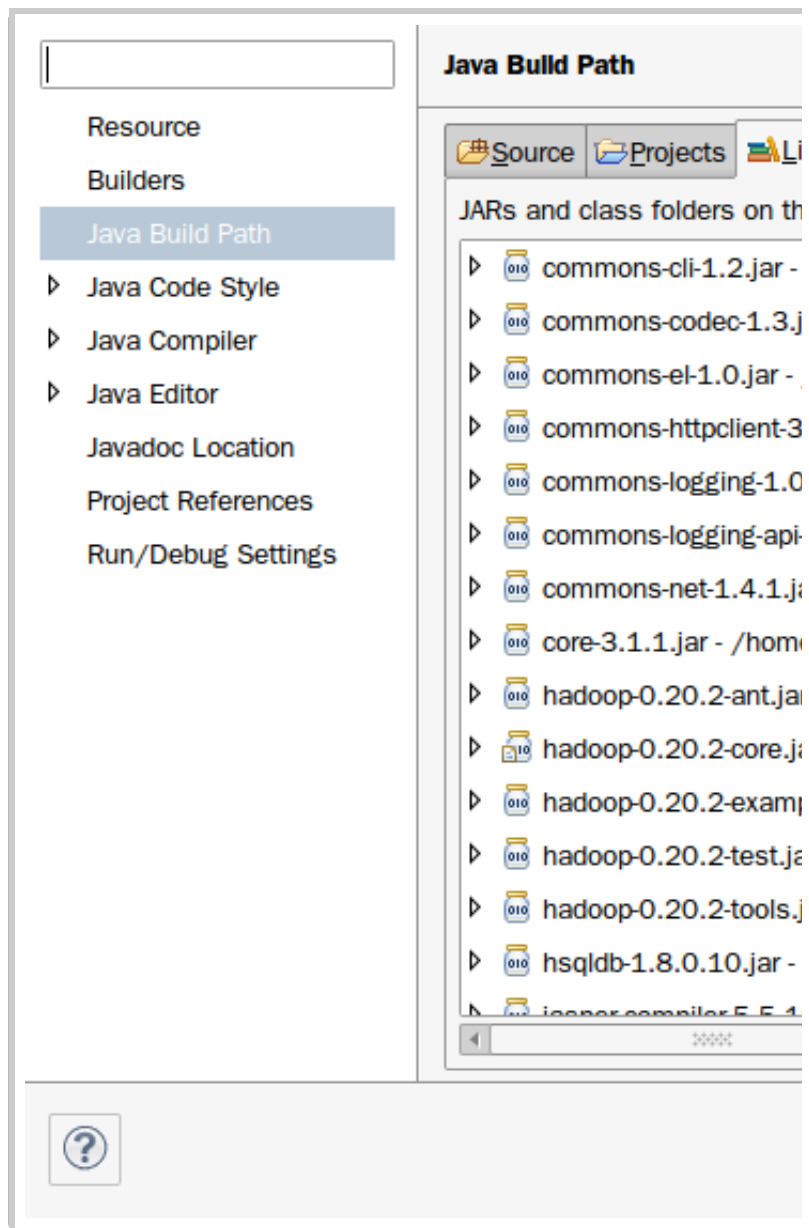
FileInputFormat.addInputPath(job,
new Path(args[0]));

FileOutputFormat.setOutputPath(job,
, new Path(args[1]));

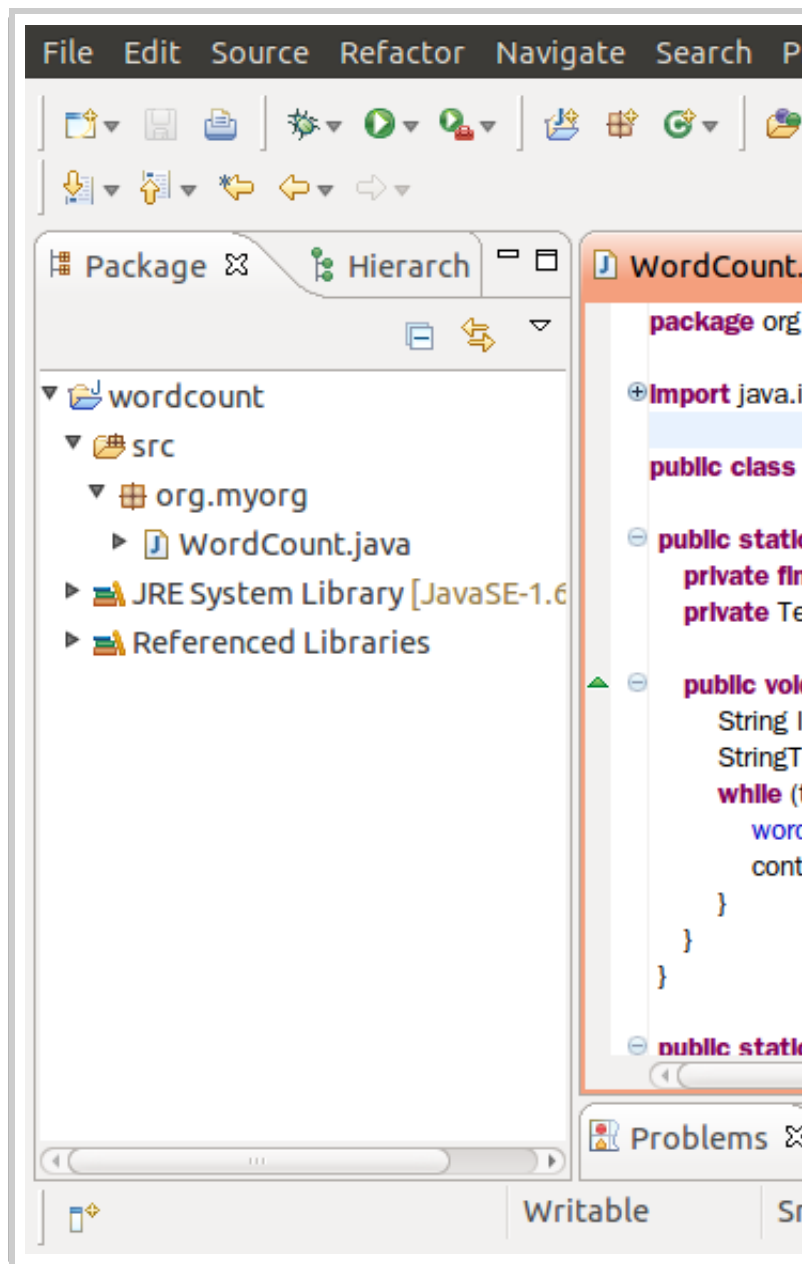
    job.waitForCompletion(true);
}

}
```

- Copy and paste the code to **WordCount.java** file and **add relevant library jar files** to your project Java Build Path.
 - In eclipse, to add jar files under HADOOP and HADOOP/lib directory, use **Project > Properties > Java Build Path > Libraries > Add External Jars**



- Finally, your eclipse will look like this:



- Export your Jar file using **File > Export > Jar file**

JAR File Specification

Define which resources should be exported into the JAR.

Select the resources to export:

☐ AmazonEM	☒ .classpath
☐ ivory_will	☒ .project
☐ MergePCPFI	
☒ WordCount	

☒ Export generated class files and resources
☐ Export all output folders for checked projects
☒ Export Java source files and resources
☐ Export refactorings for checked projects. [Select refactorings](#)

Select the export destination:

JAR file:

Options:

☒ Compress the contents of the JAR file
☒ Add directory entries
☒ Overwrite existing files without warning

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- Upload your jar file to TACC. (For instruction click [here](#))

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